

Determination of Cellular Shape by the Extracellular Matrix and Its Correlation with the Control of Cellular Growth¹

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Abstract

Although the problem of cellular proliferation may seem at first glance to be tremendously complex, the mechanisms which control it may be extremely simple. One of the primary factors which regulates the mitogenic response of a given cell type to a given class of mitogenic agents seems to be the cellular shape. We have found that corneal epithelial cells, for example, adopt a flattened configuration when maintained *in vitro* on plastic and are very sensitive to fibroblast growth factor, but not to epidermal growth factor. When maintained on collagen, on the other hand, they become tall and columnar and respond primarily to epidermal growth factor. The cellular shape is dictated *in vivo* by the extracellular material upon which the cells rest and *in vitro* by the substrate upon which the cells are maintained. The substrate itself may, in turn, induce the cells to manufacture their extracellular material and specific cell surface proteins which control the cellular shape.

Introduction

Considerable progress has been achieved in the past decade in the elucidation of the physical interaction of long-range effectors, such as steroid hormones, with their target cells. Interest has been focused primarily on the manner in which steroid hormones interact with specific receptor sites present in the cytoplasm and nuclei. It was hoped that with study of the specific interaction of steroid hormones with specific cell components, the elucidation of the basic mechanisms underlying the control of cellular proliferation would follow (2, 8, 21). Although such a relationship has been shown to exist as far as the control of cell differentiation is concerned (26), studying the specific interaction of steroid hormones with specific cell components has not proven to be a fruitful course of inquiry concerning the control of cellular proliferation. This is particularly clear in the case of one class of steroid hormones, the estrogens, which are strongly mitogenic *in vivo*, but lack mitogenic effect *in vitro*, despite the fact that they bind to specific cytoplasmic receptor sites and are subsequently transported to the cell nuclei in a way similar to that observed *in vivo*. One is therefore left with the feeling that enormous effort has been spent in trying to elucidate a physical event which ultimately has no biological effect (2). This is most regrettable, since estrogens have been shown to be involved in the control of proliferation of tissues as important for the reproduction of the individual as the

endometrium and, being that important, have been the object of considerable study for the development of anti-conception compounds. In the field of oncology, an estrogenic compound like diethylstilbestrol was shown, as early as 1932, to be responsible in part for the development of numerous hormonal cancers when derived from the kidney, pituitary, mammary gland, ovaries, or genital tract. More recently, it has been shown to be involved in the development of cancer of the vagina when administered at the fetal stage.

The lack of correlation between the binding of steroids to their target cells and their mitogenic effect *in vitro* can, in all likelihood, be traced to 2 causes: (a) the lack of an adequate *in vitro* model for the study of cell proliferation (specifically, a tissue culture model); and (b) the fact that although estrogens have been held to be preeminently responsible for the proliferation of their target cells, little concern has been given to factors which control the responsiveness of the target cells to estrogens. The lack of an adequate model *in vitro* is mostly due to the fact that estrogens act on the epithelial component of the target organ and that it is extremely difficult to imitate accurately those conditions *in vitro*, especially the cellular orientation toward the mesenchymal component as it occurs in the uterus or mammary gland *in vivo*. The problem is made no simpler by our lack of basic knowledge regarding the factors which condition cells to react to mitogens. Until that deficiency is remedied, it will be difficult to understand the biological effect of presumed mitogens such as estrogens on organs like the mammary gland or the genital tract. The same observations might be made regarding the involvement of estrogens in the early and late stages of tumor development.

Although the problem of cellular proliferation may seem at first glance to be tremendously complex, the mechanisms which control it can be extremely simple. In this communication we will provide evidence that one of the primary, if not sole, factors which regulates the mitogenic response of a given cell type to a given class of mitogenic agents is simply the cellular shape. This, in turn, could be dictated *in vivo* by the extracellular material upon which the cells rest and *in vitro* by the substrate upon which the cells are maintained; the substrate itself may in turn induce the cells to manufacture their extracellular material and specific cell surface proteins which control the cellular shape.

The Organs of Vertebrates

The organs of vertebrates consist, with but a few exceptions, of different tissues originating from the different germ layers. Those components which first develop individ-

¹ Presented at the John E. Fogarty International Center Conference on Hormones and Cancer, March 29 to 31, 1978, Bethesda, Md.

ually become associated and integrated into new organs relatively late in the course of embryonic development. It has been shown that the integration of the different tissue components of an organ are interdependent and that in the case of organs formed from a mesenchymal component, itself associated with an epithelial component derived either from the ectoderm or the endoderm, the mesenchymal component is responsible for the morphogenetic events taking place in the epithelium (16-18). This has been shown to be of particular importance in the development of tissues as diverse as the skin, kidney, liver, pancreas, salivary and mammary glands, and the urogenital tract, to cite a few examples (for a review, see Ref. 22).

Since epithelial growth is a prerequisite of morphogenesis, it is evident that the mesenchymal component of an organ can also be involved in supporting the proliferation of its epithelial counterpart. The sensitivity of a given tissue to the growth factors which govern its rate of proliferation could therefore be strongly influenced by the mesenchymal component. The role of the mesenchymal component in dictating the rate of proliferation and cytodifferentiation of its epithelial counterpart has been traced back to its ability either to secrete the basement membrane upon which the epithelial component rests (16-18) or to induce the epithelial component to secrete its own matrix or basement membrane (5, 19, 24, 25). The basement membrane will in turn be responsible for controlling the genetic expression of the epithelial component (16-19, 22). Since growth factors such as the EGF² or FGF have been involved in the control of proliferation of mammalian cells, we have attempted to determine, using the cornea and the aorta as models, whether these factors could induce the proliferation of the corneal epithelium or that of the vascular or corneal endothelium in the absence of the basement membrane upon which they rest *in vivo* or whether, to the contrary, the basement membrane is required to obtain the normal proliferative response of the epithelial or endothelial element in response to growth factors.

The Cornea

The cornea is an organ composed of 5 layers. Those layers are the epithelium, basement membrane and Bowman's membrane, stroma, Descemet's membrane, and endothelium (20). The epithelium, which is of ectodermal origin, occupies 10% of the total corneal thickness and has a high mitotic rate with a turnover of 7 days. It rests on a basement membrane, itself in close contact with the Bowman's membrane, which separates it from the mesenchymal stroma. The mesenchymal stroma is composed of fibroblasts interspersed in a very orderly and dense fibrous connective tissue. The epithelium is composed of 3 groups of cells. There is a single row of basal cells which is localized next to the basement membrane and is highly organized (Fig. 1). These cells are tall and columnar in appearance and are linked to each other by the zona occludens and adherens, which act as a sort of belt, encircling the entire cell and binding it firmly to neighboring

cells. It is in this group that the highest mitotic rate is found. Over this single layer of basal cells lies a group of winged or polygonal cells which form an intermediate zone 2 to 3 cells thick. Finally, 2 or 3 layers of flattened plate-like surface cells cover the intermediate layer (Fig. 1).

Epithelial cell layers of the cornea are known to respond to growth factors *in vivo*. The effect of EGF on the regeneration of epithelial cells of the cornea maintained in organ culture and *in vivo* was first reported independently by Frati *et al.* (7) and by Savage and Cohen (33). They demonstrated that EGF stimulated the proliferation of rabbit corneal epithelium both in organ culture and *in vivo* and that the topical application of the factor to the eyes of the rabbit with corneal wounds resulted in a marked hyperplasia of the epithelial layers, as well as acceleration of the wound healing process (7, 33).

Since epithelial cells of the cornea can also be maintained easily in tissue culture (13), *i.e.*, under conditions in which they are no longer in contact with their mesenchymal element or their basement membrane, the corneal epithelium represents an ideal cell type for comparison of the sensitivity to growth factors of cells associated or dissociated from their stroma. Furthermore, the sensitivity of corneal epithelial cells to mitogens can also be compared to that of lens epithelial cells, since both tissues arise from the same embryonic anlage (the surface ectoderm) and could be controlled by the same set of growth factors.

Sensitivity of Corneal and Lens Epithelial Cells Maintained in Tissue Culture to EGF and FGF

Individual or aggregated basal epithelial cells attached to the bottoms of plastic tissue culture dishes and formed colonies which grew as rapidly enlarging monolayers of flattened cells (Fig. 2A). Cells from bovine lens epithelium plated within 3 to 4 days and proliferated in a similar rapid manner (Fig. 2C). Careful explantation of lens or corneal epithelium resulted in virtually no keratocyte contamination, and early-passage cultures of both cell types formed compact monolayers consisting entirely of flattened epithelial cells (Fig. 2, A and C). If the corneal epithelial cells were maintained at high density without being passaged, the dish became covered with cells which exhibited a typical epithelial appearance (Fig. 2B). Transmission electron microscopy revealed that both corneal and lens epithelial cells contained numerous polysomes, as well as extensive arrays of smooth and granular endoplasmic reticula (Fig. 3). The outstanding structural features of these cells were the various forms of microfilament associations; microtubules were also present, but they were much less numerous and were not often associated with each other. Microfilaments existed in numerous closely packed bundles of various lengths and also in large sheets of loosely packed filaments (Fig. 3, B to D). In many instances the microfilaments appeared to be associated with junctional complexes (probably desmosomes) between adjacent cells (Fig. 3, B and D). These junctions were a common feature of both corneal and lens epithelial cells and were sometimes long. A particularly interesting feature was the star-like arrangement of microfilament bundles (Fig. 3C). In this arrange-

² The abbreviations used are: EGF, epidermal growth factor; FGF, fibroblast growth factor.

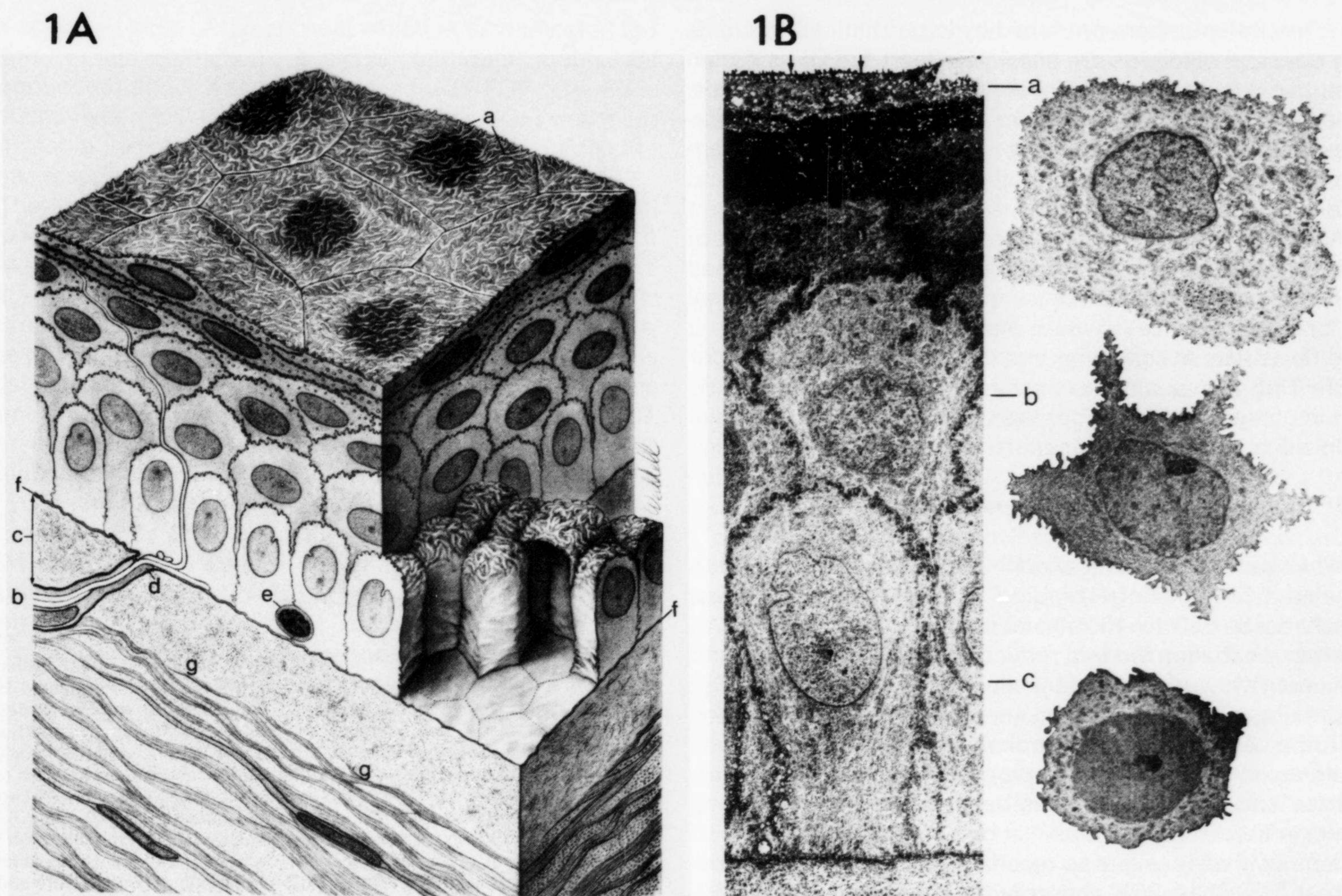


Fig. 1. A, 3-dimensional drawing of the corneal epithelium. The 5 layers of cells forming this epithelium are shown. The drawing brings out the polygonal shape of the basal and surface cells and their relative size. The wing cell processes fill the spaces formed by the dome-shaped apical surface of the basal cells. The turnover time for these cells is 7 days, and during this time the columnar basal cell is gradually transformed into a wing cell and then into a thin flat surface cell. During this transition the cytoplasm changes and the Golgi apparatus becomes more prominent. Numerous vesicles develop in the superficial wing and surface layers, and glycogen appears in the surface cells. The intercellular space separating the outermost surface cells is closed by a zonula occludens forming a barrier which prevents ingress of the precorneal tear film into the corneal stroma. The cell surface shows an extensive net of microplacae (a) and microvilli which may be involved in the retention of the precorneal film.

A corneal nerve (b) passes through Bowin's layer (c), and it loses its Schwann sheath near the basement membrane (d) of the basal epithelium. The nerve then passes between the epithelial cells toward the superficial layer as a naked nerve. A lymphocyte (e) is seen between 2 basal epithelial cells. The basement membrane is seen at (f). Some of the most superficial corneal stromal lamellae (g) are seen curving forward to merge with Bowman's layer. The regular arrangement of the corneal stromal collagen differs from the random disposition of that in Bowman's layer.

B, Corneal epithelium, electron micrograph. Note the surface microplacae and microvilli (arrows), and the vesicular structures in the cytoplasm of the outermost wing and surface cells. Desmosomal attachments between cells are seen as dark spots around the periphery of the cells. At the right are tangential sections of a surface (a), a wing (b), and a basal cell (c). Note their polygonal shape, the comparative size, and the cytoplasmic differences, especially the vesiculation in the surface layers. The cell borders show extensive interdigitations and a high density at the sites of desmosomes. $\times 4250$. Drawing and microphotograph courtesy of Dr. J. Alvarado, Department of Ophthalmology, University of California, San Francisco; originally published by W. B. Saunders & Co., New York, and reproduced here with their kind permission.

ment, several different bundles of microfilaments appeared to converge at a specific point, which could be a desmosome, to form a dense mass of filaments (13).

Effects of FGF and EGF on [3 H]Thymidine Incorporation and Proliferation of Lens and Corneal Cells Maintained in Tissue Culture

FGF and EGF were first tested on bovine corneal and lens epithelial cells by examining the DNA labeling indices in quiescent cell cultures after the addition of mitogens. Fig. 4 shows that control cultures of corneal epithelial cells had a labeling index of less than 10%, while cultures to which optimal levels of serum or FGF had been added showed labeling indices of up to 90% (Fig. 4, B and C). However, addition of EGF had no effect on [3 H]thymidine incorpora-

tion in these cultures; labeling indices of cultures containing EGF were not significantly greater than were those of control cultures (Fig. 4D). The same results were obtained with similarly treated lens epithelial cell cultures (13). Thus FGF, like calf serum, stimulated the initiation of DNA synthesis in resting epithelial cells in tissue culture, while EGF did not produce this effect.

These effects of FGF and EGF on [3 H]thymidine incorporation were substantiated by examining cell proliferation in the presence of these mitogens. The proliferative response of corneal and lens epithelial cells to increasing concentrations of FGF and EGF was examined to find the minimal and optimal concentrations of FGF which would stimulate cell proliferation and to see whether EGF would be mitogenic for these cells. Cells were cultured for 7 days in media containing various levels of FGF or EGF and then counted.

The final cell numbers produced by logarithmically increasing doses of mitogens are shown in Chart 1. EGF is not an effective growth promoter for corneal or lens epithelial cells at concentrations ranging from 10^{-3} to 10^3 ng/ml. These results demonstrate that when corneal epithelial cells are maintained in tissue culture, they no longer respond to EGF.

A number of explanations may be offered for the lack of effect of EGF on bovine corneal epithelial cells maintained in tissue culture. One of the simplest could be that corneal or lens epithelial cells, when maintained in tissue culture, dedifferentiate in some way and lose their receptor sites for EGF. That this was not the case was demonstrated by the binding capacity for EGF of the corneal and lens epithelial cells maintained in tissue culture.

EGF Binding to Corneal and Lens Epithelial Cells

When cultures of corneal epithelial cells were exposed to labeled EGF, maximal binding of 125 I-labeled EGF was reached after 30 to 40 min incubation at 37° (Chart 2A). Further incubation led to a reduction in the total cell-bound radioactivity until a constant amount of 125 I-labeled EGF, approximately 40% below maximum, remained associated with the cells. It is thereby indicated that, after EGF binds to its receptor sites, the receptor site complexes are internalized and degraded, as has been observed with human foreskin fibroblasts (4). Maximal binding of 125 I-labeled EGF to lens epithelial cells was reached after 60 min of incubation at 37° (Chart 2C), and continued incubation with 125 I-labeled EGF did not give rise to a reduction in cell-bound radioactivity. 125 I-labeled EGF bound to either corneal epithelial or lens epithelial cells was displaced by the addition of unlabeled EGF (500 ng/ml) to the binding medium (Chart 2, A and C).

The effect of the EGF concentration on binding was studied by measuring the maximal binding after the addition of increasing quantities of 125 I-labeled EGF to the incubation medium. Scatchard plots of these data are

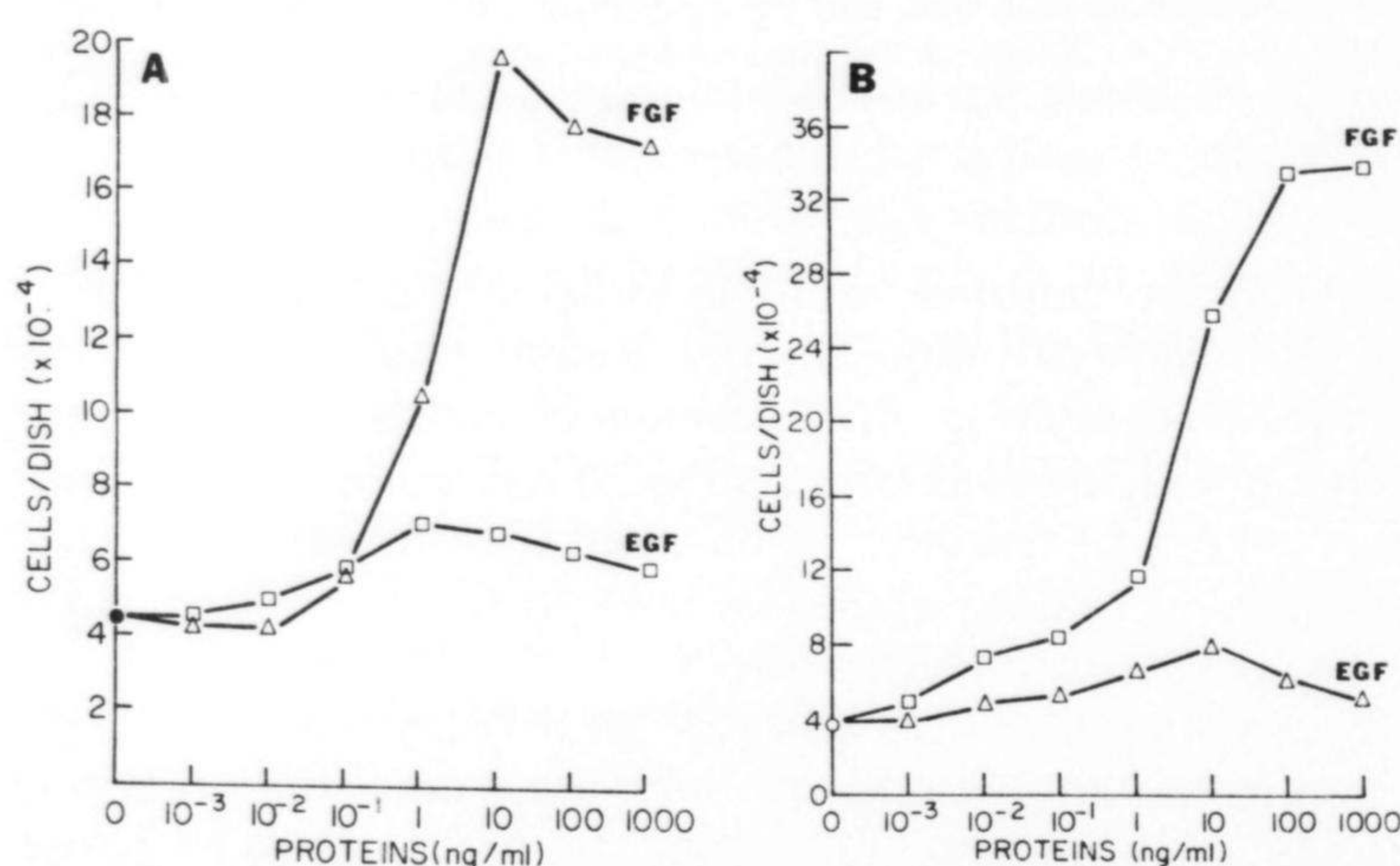


Chart 1. Effects of increasing concentrations of FGF and EGF on proliferation of epithelial cells. The cells were plated as described in Ref. 12 at densities of 30,000/dish for the corneal epithelial cells (A) and 18,000/dish for the lens epithelial cells (B) and maintained in the presence of 0.5% calf serum, with increasing concentrations of FGF or EGF. Media were changed every 2 days, and on Day 7 all cultures were counted. Each symbol represents the mean of duplicate cultures; S.D. did not exceed 10% of the mean.

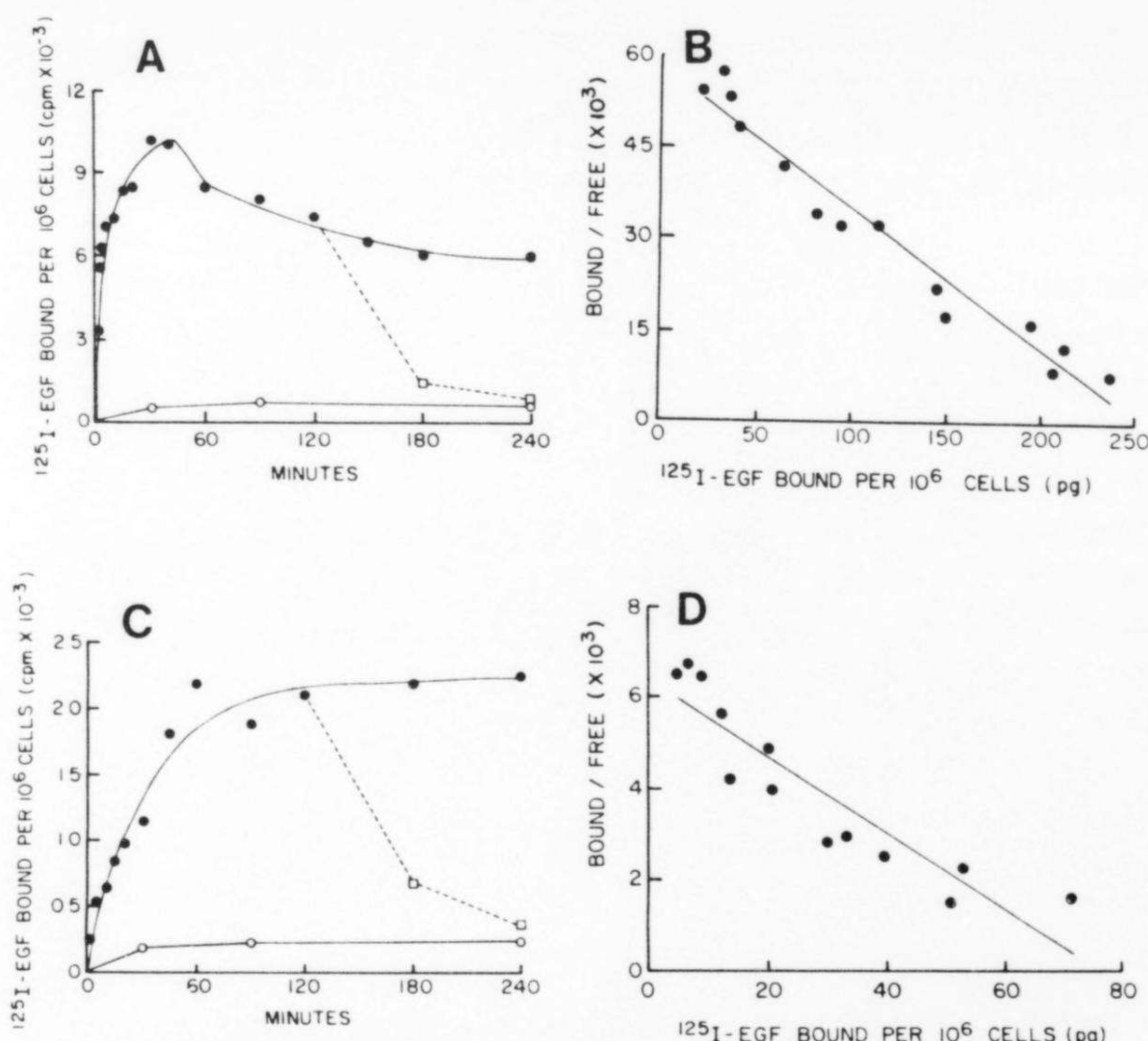


Chart 2. A, time-course of 125 I-labeled EGF binding to corneal epithelial cells. Two ml of binding medium, containing 6 ng/ml of 125 I-labeled EGF (45,000 cpm/ng), were added to each dish, and the dishes were incubated at 37° . At the indicated time interval, the cell-bound radioactivity ($\bullet$) was determined as described in Ref. 12. Nonspecific binding was measured by adding 500 ng/ml of unlabeled EGF to the incubation medium at 0 min ($\circ$). Displacement of bound EGF from the cells was followed after the addition of 500 ng/ml of unlabeled EGF to the incubation medium at 120 min ($\square$). B, Scatchard plot showing the effect of 125 I-labeled EGF concentration on binding to corneal epithelial cells. Various concentrations of 125 I-labeled EGF (0.2 to 30 ng/ml, 40,000 cpm/ng) in 2 ml of binding medium were added to the dishes, and the cell-bound radioactivity was determined after 40 min of incubation at 37° , as described in Ref. 12. C, time course of 125 I-labeled EGF binding to lens epithelial cells. The specific activity of the labeled EGF in this experiment was 65,000 cpm/ng. Other experimental details are identical to those above. D, Scatchard plot showing the effect of 125 I-labeled EGF concentration on binding to lens epithelial cells. Various concentrations of 125 I-labeled EGF (0.5 to 45 ng/ml, 65,000 cpm/ng) in 2 ml of binding medium were added to the dishes, and the cell-bound radioactivity was determined after 60 min of incubation, as described in Ref. 12.

shown for corneal epithelial cells (Chart 2D) and lens epithelial cells (Chart 2E). The binding reaction was shown to be a saturable process with half-maximal binding at 4:3 ng/ml (7×10^{-10} M) for the corneal epithelial cells and 11:8 ng/ml (1.9×10^{-4} M) for the lens epithelial cells. The total binding capacity of corneal epithelial and lens epithelial cells was found to be 250 and 75 pg per 10^6 cells, respectively. The passage number of the cultures did not affect the binding of EGF to the cells, since it was similar in early- and in late-passage cultures. It was therefore demonstrated that although the lens and corneal epithelial cells exhibit specific receptor sites for EGF, they will not respond to the mitogen when maintained in tissue culture.

Effects of EGF and FGF on Corneal Epithelium in Organ Culture

Corneas from bovine fetuses were maintained in organ culture for 5 days in the presence or absence of 1 μ g/ml of EGF or FGF. Fig. 5 shows the morphology of the tissues after 5 days in culture. As already reported by Savage and Cohen (33), the addition of EGF to the culture medium resulted in a marked enhancement of the proliferation of

the epithelial cells of the cornea, which in turn affected the entire corneal epithelium. A similar (although much less pronounced) effect was also achieved with FGF. The FGF-induced hyperplasia, unlike that induced by EGF, was most noticeable in specific areas of the cornea, while adjacent areas responded poorly or not at all.

The study of the mitogenic sensitivity of bovine corneal and lens epithelium to FGF and EGF leads to the conclusion that although corneal epithelial cells in tissue culture are responsive to FGF and not to EGF, in organ culture EGF is a strong mitogen for the corneal epithelium, while FGF is less potent. One might wonder, therefore, what differences exist between the 2 culture conditions which make the corneal epithelium sensitive to the mitogenic effect of EGF. The main difference between the corneal epithelial cells maintained in tissue culture (where the cells are insensitive to EGF) and those maintained in organ culture (where they are sensitive to EGF) is in the substrate upon which the cells rest. In tissue culture, the epithelial cells proliferated on plastic as a collection of individual cells which had lost their orientation to each other and had a flattened appearance. In organ culture, the integrity of the corneal epithelium is preserved, and the topological relationships that exist *in vivo* are present between the stroma (of mesenchymal origin), the basement membrane, and the epithelium, as is the topological relationship between the different layers of the epithelium (columnar basal cells *versus* polygonal wing cells).

The Role of the Stroma in Inducing the Corneal Epithelial Cells to Respond to EGF

The basal cell layer of the corneal epithelium rests *in vivo* on the Bowman's membrane, which is composed of collagen associated with proteoglycans. The Bowman's membrane itself is in direct contact with the corneal stroma, which is composed of highly organized layers of collagen, among which the keratocytes are interspersed. The inductive effect of the stroma on the proliferation of the corneal epithelial cells maintained in organ culture could then be due either to the presence of keratocytes or to the possible involvement of collagen and other components of the extracellular matrix. Since it has been shown by others that the epidermal cells of the skin proliferate and differentiate in response to EGF when they are maintained on collagen (23) or on a feeder layer of fibroblasts (31), the effect of collagen and that of fibroblasts on the mitogenic response of corneal epithelial cells to EGF were compared.

The Role of Collagen as a Substrate for Growing Epithelial Cells, as Compared to Plastic and Feeder Layers of Fibroblasts. When corneal epithelial cells were plated on collagen-coated dishes (Fig. 6D), they reached a higher cell density than when they were maintained on plastic (Fig. 6A), and the addition of FGF to the culture did not result in any marked increase in the rate of cell proliferation (Fig. 6F). In contrast, the cultures maintained on collagen-coated dishes responded to EGF with both a marked increase in cell number (Fig. 6C) and an increased rate of keratinization which reflected the increase in cell number.

It was also observed that a noticeable change in the

shape of the cells occurred when they were maintained on collagen as opposed to plastic. On plastic, the cells remained flattened (Figs. 2B and 7A), while on collagen they were rounded (Fig. 7B) and began to proliferate actively during their first week in culture (Fig. 7C). They formed a monolayer of packed, tall, and columnar cells (Fig. 7C) and began to form multiple layers during the second week, with the cells of the second and third layers having the appearance of winged cells (Fig. 7D). During the third week, keratinization occurred in the upper cell layers, as indicated by the appearance of keratinized cellular envelopes (Fig. 7E) and positive histochemical staining with rhodamine B (Fig. 7F). Basal cells maintained on collagen can therefore mimic the complete sequence of events present in the corneal epithelium (differentiation into winged and plated cells), while they do not progress beyond the stage of a monolayer when maintained on plastic.

Similar (although much less marked) differences were obtained when corneal epithelial cells were seeded on a feeder layer of Swiss 3T3 cells (Fig. 6G), instead of on collagen. These results are consistent with the observations of Sun and Green (34), who have shown that corneal epithelial cells maintained on feeder layers of Swiss 3T3 cells respond to EGF with an increased rate of proliferation. Therefore, these results demonstrate that in the case of an organ such as the cornea, which is comprised of a mesenchymal component and an epithelial component, the mesenchymal component can be held responsible for the proliferative activity of the epithelial part, since the main factor determining whether epithelial cells will divide in response to EGF is whether or not they rest upon the stroma. Moreover, since corneal epithelial cells will proliferate equally well when maintained either on collagen or on a feeder layer of fibroblasts, it is probable that the primary factor determining the proliferation of basal cells is whether or not the cells are in contact with the collagenous matrix which can be produced *in vivo* either by the mesenchymal element or by the corneal epithelium itself, when the latter is maintained under the proper conditions (9). Dodson and Hay (5) have demonstrated that isolated corneal epithelium grown on killed lens capsule produces a collagenous stroma *in vitro* which is reminiscent of that formed *in vivo*, while Meier and Hay (23) have reported that any type of collagenous substratum tested supports the efforts of the corneal epithelium to produce a stroma. On glass, plastic, and substrates of noncollagenous proteins, however, the corneal epithelium fails to produce a stroma.

The Vascular and Corneal Endothelia

The vascular and corneal endothelia constitute the inner lining of the blood vascular system and cornea, respectively. They consist of a monolayer of highly flattened cells which are highly contact inhibited and are respectively exposed on one side to the blood or the aqueous humor and on the other side to the vessel media or the corneal stroma. They therefore exhibit both a morphological and functional polarity. The side exposed to the bloodstream is a nonthrombogenic surface, while the side exposed to the stroma is thrombogenic. It is on this thrombogenic surface

that the synthesis of the respective basement membranes (basal lamina for the vascular system and Descemet's membrane for the corneal endothelium) occurs. The physiological function of the vascular and corneal endothelium is that of a selective barrier capable of active transport with a given polarity. In the vascular system, transport proceeds from the bloodstream toward surrounding tissue, while in the cornea it proceeds from the stroma toward the aqueous humor.

The Corneal and Vascular Endothelia and Their Basement Membranes *in Vivo*. The corneal and vascular endothelial cells can easily be maintained in tissue culture, provided that FGF, which has been shown to be a mitogen for these cell types (9-12, 14, 15), is included in the tissue culture medium (Chart 3). When the cells reach confluence, the cultures exhibit a morphological appearance similar to that observed *in vivo* (Figs. 8 and 9). This is concomitant with the formation in either culture of a basement membrane which has characteristics similar to those observed *in vivo*.

When corneal or vascular endothelial cell cultures are treated with detergents such as Triton X to remove the cell layer, a basement membrane composed of multiple layers of collagenous material is exposed (Figs. 10 and 11). This basement membrane is itself coated with fibronectin (Figs. 11 and 12), which has been shown by others to be responsible for both cell adhesion and contact inhibition. Furthermore, the corneal and vascular endothelial cells maintained in tissue culture will secrete their basement membranes and the associated fibronectin only on the endothelial cell surface which is exposed to the substrate, not on the endothelial surface exposed to the medium (3, 10). Thus, the polarity observed *in vivo* is respected *in vitro* (Fig. 13).

One of the main differences, therefore, between vascular or corneal endothelial cells and corneal epithelial cells maintained *in vitro* is the ability of the former to secrete a

basement membrane. While the vascular and corneal endothelia retain this ability when maintained on plastic, the corneal epithelial cells are no longer capable of producing their basement membrane and thereby exhibit a requirement for either collagen-coated dishes or for feeder layers when they are to exhibit their normal morphology.

The Control of the Proliferation of Vascular and Corneal Endothelial Cells. When the growth response of bovine vascular endothelial cells to FGF and EGF was compared, it was observed that regardless of the vascular territory of origin, the flattened vascular endothelial cells always respond to FGF, but never to EGF. With human umbilical vascular endothelial cells which, in contrast to bovine vascular endothelial cells, have receptor sites for EGF (9), the mitogenic response to EGF was one-tenth that of cultures exposed to FGF alone (9, 10). This demonstrated that FGF was the prime mitogen for vascular endothelial cells, which in culture adopt a flattened morphology similar to that observed *in vivo*. Similar results were obtained with corneal endothelial cells. Although these cells respond to both EGF and FGF, FGF was in every case a far stronger mitogen than was EGF (11). One could therefore conclude that for the corneal epithelial cells which are maintained on plastic and have a flattened morphology, FGF is the prime mitogen. One should observe, however, that the flattened morphology is the characteristic *in vivo* morphology of vascular and corneal endothelial cells, while in corneal epithelial cells it is an *in vivo* aberration. Since corneal epithelial cells become preferentially sensitive to EGF when they become tall and columnar, we may conclude that cellular morphology dictates the sensitivity to growth factors. The tall, columnar basal epithelial cells found in the skin, bladder, and digestive tract, to cite a few examples, should be preferentially sensitive to EGF, while the highly flattened vascular and corneal endothelial cells, as well as the mesothelial cells, should be sensitive primarily to FGF. If such is the situation, one may wonder what the extracellular signals are which will dictate the shape of the cells and how they could, in turn, influence the cytoskeleton of the cells, which is primarily responsible for the morphological appearance of a given cell type.

The Nature of the Inductive Process Involved in the Modulation of Epithelial and Endothelial Cell Proliferation and Cytodifferentiation

The nature and number of interacting factors involved in a given inductive process are unknown and can vary from organ to organ. Most evidence now favors the theory that they are steric, in which case the extracellular matrix will allow the cells to adopt a configuration such that the mitogen can interact directly with the organized epithelial layers or with the endothelial monolayer to induce proliferation of the basal columnar epithelial cells at all times or of the vascular or corneal endothelial cells in cases in which the cells are released from contact inhibition (wounds).

The Steric Effect of the Extracellular Matrix. The substrate upon which the cells rest will dictate the cellular morphology, and cellular morphology can in turn determine the sensitivity of cells to mitogens. That both the shape of a

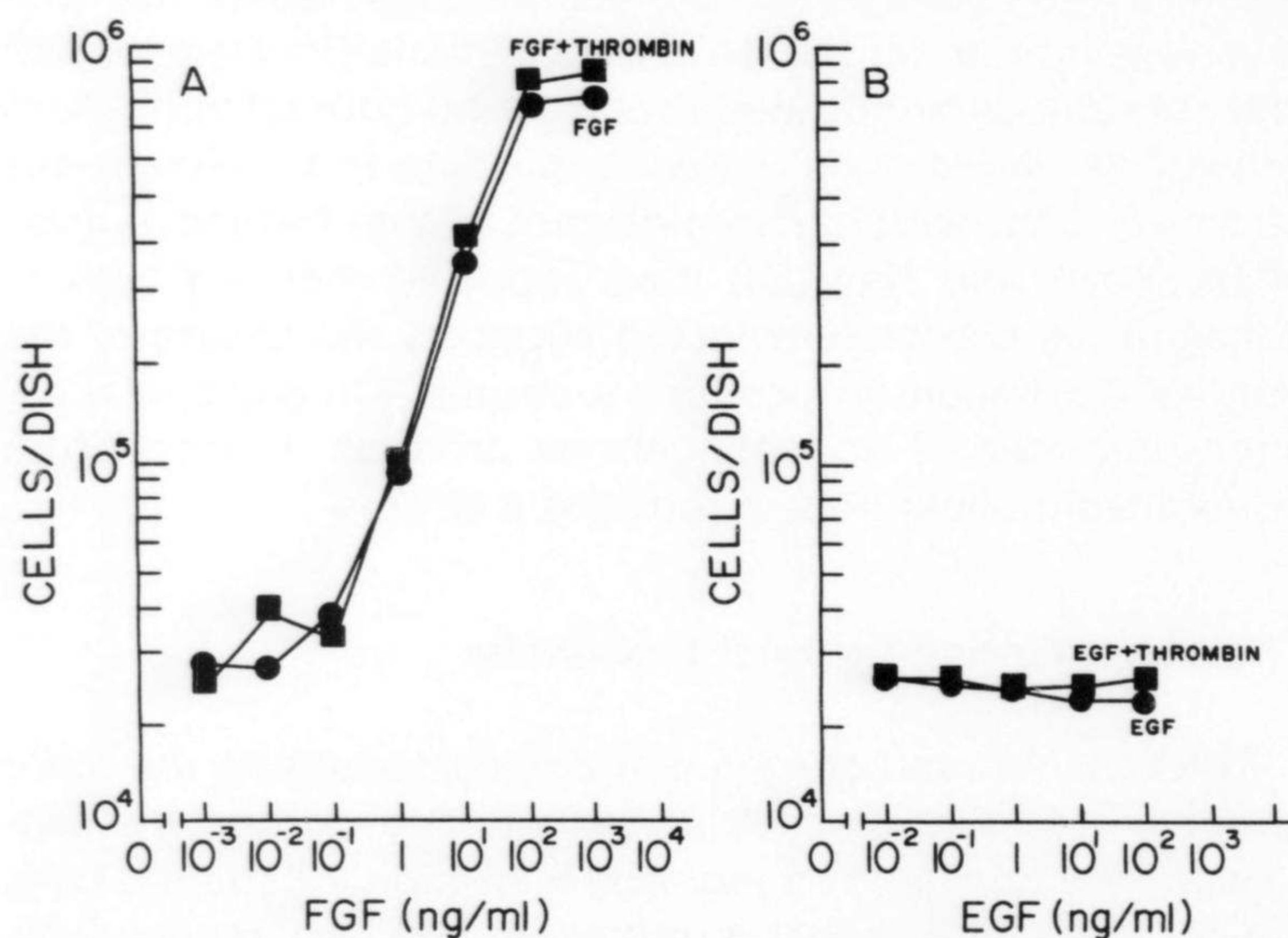


Chart 3. Effects of EGF, FGF, and thrombin on the rate of proliferation of bovine endothelial cells. Cells were plated at 370 cells/sq cm in 6-cm culture dishes in Dulbecco's modified Eagle's medium containing 10% calf serum. FGF (A) and EGF (B) were added at the indicated concentrations every other day, and the medium was renewed on Day 4. After 7 days triplicate cultures were trypsinized and counted. Controls treated with only thrombin contained 3.6×10^4 cells.

cell and its orientation *vis-à-vis* neighboring cells are important factors in modulating the proliferative response of a given cell type to a mitogen is by no means a new observation. As early as 1964, Wessells (36), using as a model the basal layer of chick embryonic epidermis, which is normally columnar and mitotically active, demonstrated that a loss of mitotic activity resulted when the dermis and epidermis were separated in such a way that their previous orientation to each other was destroyed. In such a case the cells also became flattened and lost their columnar appearance. If, on the other hand, the cells were maintained on the proper substrate (in this case refrozen thawed dermis, tropocollagen, or Millipore filters) and exposed to chick embryo extract, the cells remained columnar and mitotically active. This indicates that the orientation of the cells, their shape, and their proliferative activity are closely linked properties of the basal germinal population. Similarly, it is likely that in the case of the corneal epithelium the proliferative activity of the basal layer will be linked to its columnar morphology. If the cells flatten, as they do when maintained in tissue culture on a substrate such as plastic, they will lose their responsiveness to growth factors to which they normally respond *in vivo*, while developing sensitivity to growth factors to which they do not respond under physiological circumstances. In the case of the corneal epithelium, therefore, the extracellular matrix upon which it rests will influence both its proliferative rate and its cytodifferentiation.

The Factors Involved in Dictating Cellular Shape: the Cell Surface Proteins and the Microfibrillar System. The extracellular matrix associated with a given tissue is composed mainly of collagen associated to small amounts of proteoglycan and, in all likelihood, to numerous other compounds yet to be defined and isolated. It has, for example, only recently been demonstrated that fibronectin, one of the major cell surface proteins, is found in most basement membranes. In this case, one wonders whether it is a cell surface protein or a basement membrane protein. Fibronectin has been shown to be involved in cellular adhesion and seems to function in most tissues as a glue by simply reinforcing the adhesiveness of the cell to its basement membrane. In some tissues, such as the vascular and corneal endothelia, it could have a far more important role, in view of the morphological characteristics of those tissues. The vascular and corneal endothelia are highly contact-inhibited tissues in which overgrowth brings catastrophic results. In the cornea, this is reflected by Fuch's dystrophies, a condition in which local overgrowth of the corneal endothelium gives rise to a loss of endothelial function followed by loss of vision. In the vascular system, overgrowth, in view of the nonthrombogenic surface of the vascular endothelium and the narrowing of the blood vessels, will very probably lead to local ulceration processes and thrombosis.

Fibronectin has been shown to be involved in controlling cellular morphology. This is particularly true of transformed cells, for which addition of fibronectin to the tissue culture medium partially restores the normal morphology, adhesiveness, and contact inhibition of movement (37). With respect to the vascular and corneal endothelia then, fibronectin, because of its preferential accumulation in the base-

ment membrane of those tissues, could control and/or mediate the contact-inhibited appearance of those tissues (3, 10). This morphology will in turn limit the proliferative potential of the vascular and corneal endothelia since, in contrast to the corneal epithelial cells, they do not progress beyond the monolayer stage. Furthermore, the massive accumulation of fibronectin on the cell surface exposed to the substrate may result in a restriction of the lateral mobility of various cell surface receptor sites within the plasma membrane and thereby may dictate the physiological function of the vascular and corneal endothelia, which is that of a selective barrier with a polarity of transport (3, 10). One might then conclude that in the vascular and corneal systems, fibronectin present in the extracellular matrix could have an important role in cell-cell and cell substrate interactions and thereby may affect cellular growth characteristics, metabolic functions, and cytodifferentiation.

What, then, dictates the morphological appearance of the basal cell layer of the corneal epithelium?

The extracellular matrix of the corneal epithelium is composed of collagen associated to small amounts of proteoglycan. While the collagen itself may not have any direct morphogenetic role, it is nevertheless likely that when it is used as a substrate for the cultivation of corneal epithelial cells, the cells respond by forming specific cell surface proteins which could in turn induce the cells to aggregate and adopt the configuration of a columnar monolayer. Such a hypothesis finds support when one compares the behavior of other cell types, such as corneal or vascular endothelial cells which, when maintained on the proper substrate, are capable of producing the fibronectin.

An easy way to test this hypothesis is radioiodination labeling of the cell surface proteins of basal corneal epithelial cells maintained on plastic, instead of on collagen-coated, dishes and to then compare their distribution on acrylamide gel. When cells maintained on collagen acquire the ability to synthesize specific cell surface membrane proteins which could then be held specifically responsible for the morphological appearance of the cells, they should be readily detectable.

The next question is how those cell surface proteins could influence the shape of the cells. Although this is a difficult question to answer, they may well influence the shape of the cells by influencing the distribution of the microfibrillar system. While in cultured vascular endothelial cells the microfibrillar structure runs parallel to the plasma membrane (Fig. 13), similarly, the microfibrillar structure in corneal endothelial cells is found in a zone free of organelles just below the plasma membrane of the cells. It runs in a plane parallel to the cell membrane and originates from the terminal web found on the lateral side of the cells (20). In epithelial cells, especially those of columnar appearance, the microfibrillar structure acquires a degree of complexity which is reflected in the star-like distribution observed with lens epithelium in particular (Fig. 3C). One is therefore left with the impression that in flattened cells the microfibrillar system is parallel to the plasma membrane and thus hardens the plane of the cells, while in tall, columnar cells it will be organized in bundles perpendicular to the plasma membrane and originating from desmosomes, thereby acting as an intrastucture to maintain their columnar shape.

Cell Shape and Cell Proliferation. That cellular shape is an important factor in the regulation of the sensitivity of cells to mitogens becomes evident when one considers the case of cells where the proliferative rate is anchorage dependent (for a review, see Ref. 5). When diploid cells such as fibroblasts are plated on glass or plastic, they will spread on the surface and proliferate actively in response to serum factors, until they reach confluence. If, however, the same cells are prevented from attaching and remain rounded, they will no longer divide in response to serum factors, although they survive for prolonged periods. The behavior of a subcell line of the L-6-myoblast line which was conditioned to grow either in suspension or when attached to the substrate (35) had led us to the conclusion that the growth factors which govern the proliferation of cells which have a "spread" configuration may be different from those governing cells with a rounded configuration. While the subcell line mentioned above responded to both FGF and serum factors with an increased rate of proliferation when grown in the attached configuration, when grown in suspension it no longer responded to FGF, although it did retain its sensitivity to serum factors (35). It is thereby demonstrated that the shift of cellular configuration induced a loss of sensitivity to FGF, but not to serum factors. Since, moreover, the serum contained more than one type of growth factor, it seems likely that the serum mitogen which triggered the attached L-6 cell line to divide is not the same one which triggered the unattached and rounded cells to do so. Further evidence that the substrate upon which the cells grow can dictate their response to growth factors is provided by the Chinese hamster ovary cell line which, although it does not respond to insulin when grown on plastic, will respond to insulin present at physiological doses when maintained on agar (1).

It should be pointed out that transformed cells have escaped the restrictions imposed on cell proliferation by cellular morphology. Cell transformation is, in most cases, accompanied by a loss of anchorage dependence, a lower requirement for growth factors, and the acquisition by the transformed cell of the ability to grow in suspension (for a review, see Ref. 32). It is thus suggested that in the case of transformed cells there is no longer any correlation between cell shape and the control of cell proliferation.

The Substrate and Its Genetic Information. The substrate upon which the cell rests is not important merely in determining the cell's shape and its growth. It has also been shown to be of extreme importance in inducing cells to shift their pattern of differentiation, thereby modifying their genetic expression. Proof that this is the case is provided by the impressive work of Reddi et al. (27-30). These authors have reported that demineralized bone matrix implanted in s.c. sites can induce the entire endochondrial sequence. Within 3 days, fibroblasts migrate on the collagenous surface, and following migration, they proliferate actively and give rise to cartilaginous tissue. By Day 9 the chondroblasts have become chondrocytes and begin to hypertrophy. As capillaries begin to sprout, chondrolysis becomes evident. By Day 12 new bone begins to form and chondrolysis is complete. Hemopoiesis is, at this point, near culmination, and the formation of bone marrow is

completed by Day 24. Therefore, an inert matrix-like demineralized bone is capable of inducing in fibroblasts a completely different genetic expression, which will lead to the formation of cartilaginous bone and bone marrow in a time sequence reminiscent of the embryonic formation of these tissues. This demonstrates that a given substrate not only controls the differentiation of the cells normally associated with it, but also can shift the pattern of genetic expression of another cell type already differentiated, as in the case of fibroblasts. Similar effects of the matrix could also occur during the embryonic development, when extensive migration of cells which later form diverse organs is observed.

Conclusion

Although tissue culture techniques have been widely used in the study of cell migration, proliferation, and differentiation, they have their own limitations, which have, in some cases, not been fully recognized. One of those limitations is exemplified by the cornea. In contrast with the conditions which prevail *in vivo* and in organ culture, wherein the topological relationships which exist between stroma, Bowman's membrane, basement membrane, and the epithelium, as well as the topological relationship among the different layers of the epithelium, are maintained, in tissue culture the epithelial cells proliferate as a collection of individual cells. The second limitation is that, while organ culture is favorable to growth in 3 dimensions, tissue culture favors growth in 2 dimensions. Therefore, in the case of skin, corneal, and intestinal tract epithelia, to cite a few examples, the tissue culture technique does not seem appropriate. In contrast, endothelial cells, which grow *in vivo* in 2 dimensions, seem to adapt quite readily to tissue culture conditions, imitating exactly their *in vivo* condition, even to the extent of making their own basement membrane similar in composition to that seen *in vivo*. Therefore, studies dealing with endothelial cells maintained in tissue culture are likely to be more representative of the *in vivo* situation than are studies dealing with epithelial cells maintained in tissue culture.

Parallel to the lack of differentiation of epithelial cells in tissue culture is their inability to secrete a basement membrane. Such cultures must therefore be grown either on collagen or on feeder layers of fibroblasts which will make the collagen. Maintenance on such a substrate will in turn favor the synthesis of extracellular material by the cells themselves. As has been recently demonstrated in transfilter experiments (5, 19), it is only when all these conditions prevail that the cells will behave as *in vivo*. The most noticeable difference between endothelial and epithelial cells maintained in tissue culture resides in the ability of endothelial cells, but not of epithelial cells, to make their own basement membrane. Since this function seems to correlate with the capacity or incapacity of the cells to differentiate *in vitro*, one may conclude, as others have (16-18, 36), that the basement membrane is among the essential components which induce cell proliferation, cytodifferentiation, and morphogenesis of a given tissue.

The use of cell types which do not have the ability to make their own basement membrane when maintained in

tissue culture should therefore be looked upon with caution when studying the proliferation of cells and their response to mitogenic agents. If one takes the cornea as a model, it is found that EGF is mitogenic for the corneal epithelium both *in vivo* and *in vitro*, provided that the corneas are maintained in organ culture. Homogenous populations of corneal epithelium maintained as tissue cultures *in vitro*, on the other hand, fail to proliferate in response to EGF. Conversely, serum and plasma are mitogenic for the corneal epithelial cells maintained in tissue culture, but not for those maintained in organ culture. This indicates that the use of corneal epithelial tissue cultures as a bioassay for the mitogenic activity of factors purified from plasma or serum will selectively isolate factors that are active in tissue culture, but are probably inactive either in organ culture or *in vivo*. For example, although the corneal epithelium is responsive to EGF, any attempt to purify EGF from tissue by monitoring the extract for an effect on corneal epithelium tissue cultures would, of course, fail. The use of organ cultures, however, would result in the purification of EGF. This demonstrates that a single cell type, maintained *in vitro* under 2 different culture conditions, could give rise to the isolation of 2 completely different mitogenic agents. Of these 2, the one isolated by use of the organ culture assay would be most likely to have activity *in vivo*.

Although the tissue under consideration was the corneal epithelium, one could wonder whether the same conclusions do not apply to other epithelia as well, particularly in the case of tissues such as the endometrium, upon which estrogens have no mitogenic effect *in vitro*, although they have been shown to be potent mitogenic agents for these tissues *in vitro*.

The substrate for cells growing in 3 dimensions is a major factor in the control of cell shape, and it thereby indirectly controls the sensitivity of the cells to mitogens. One can therefore envision a situation in which, with local alteration of composition of the substrate, cells resting on it will be induced to respond to mitogens to which they were previously unresponsive. This could have far-reaching implications for the development of localized cell proliferation in a given tissue. Let us take the epithelium as an example. In its normal composition, the basal cell layer is sensitive to EGF, but not to FGF. When the concentration of EGF to which the tissue is exposed varies with time, but the concentration of FGF is kept constant and saturating, the cells, because of their shape, will never respond to FGF, but will respond to EGF, provided that it is present at times in sufficient concentration. A burst of mitotic activity in the epithelium will result which can be modulated by altering the local concentration of the mitogen and, although the entire basal layer will proliferate, it will do so only some of the time, not most of the time. If one interferes with normal cell function in a localized area of the epithelium in such a way that the basement membrane is altered and a flattening of the cells results, the cells in that localized area will regain their sensitivity to FGF, while losing their responsiveness to EGF. Since the concentration of FGF is saturating, continuous proliferation will result. In contrast with the first situation, however, not all of the cells of the basal layer will proliferate; only those present in the localized area where the basement membrane has been altered will proliferate.

Continuous and localized cell proliferation will therefore result where previously only occasional proliferation was observed.

Similar results can be achieved by providing the cells with a foreign substrate upon which the cells can migrate. Glass, plastic, or steel have all been shown to increase the incidence of sarcoma, while asbestos fiber will induce mesoendothelioma (for review, see Ref. 6). One of the early steps of cell transformation may therefore be induced either by altering the basement membrane or by introducing a foreign substrate which will alter the shape of the cells.

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Fig. 2. A, monolayer of bovine corneal epithelial cells maintained in Dulbecco-Vogt modified Eagle's minimum essential medium (DME) supplemented with 10% calf serum. Phase-contrast, $\times 150$. B, monolayer of basal bovine corneal epithelial cells maintained in tissue culture for over 3 weeks; areas of differentiation are visible. C, monolayer of bovine lens epithelial cells maintained in DME supplemented with 10% calf serum. Phase-contrast, $\times 150$. Lenses were removed from the eyes, and fragments from the periphery of each lens were removed with sterile forceps or a cotton-tipped applicator and placed in 10-cm dishes containing 15 ml of DME with 10% calf serum. When the epithelial fragments had attached to the bottoms of the dishes, the medium was replaced with fresh DME containing 10% calf serum and changed twice a week thereafter.

Freshly enucleated bovine eyes were collected from the local slaughterhouse. The corneas were rinsed thoroughly with sterile phosphate-buffered saline, and the epithelial layer with Bowman's membrane was separated from the stroma. The tissue was placed stroma side down in a tissue culture dish and incubated in 0.5% collagenase at 37°. After 1 hr the tissue was rinsed several times and incubated in a 0.25% trypsin-0.002 M EDTA solution at 37° for 3.5 hr. The trypsin solution was renewed every hr, at which time the loose layers of epithelial cells were removed. By 3 hr, only a single sheet of basal epithelial cells remained and detached from Bowman's membrane after an additional one-half hr. The cells were collected in 10% calf serum and DME, spun, resuspended in the same medium, and plated in tissue culture dishes (plastic substrate).

Fig. 3. Electron microscopy of thin sections of cultured bovine lens epithelial cells. Arrows, specialized junctions, possibly desmosomes, between adjacent cells. Note the star-like arrangement of microfilament bundles in C. m, mitochondria; ef, extracellular fibrous material, possibly collagen. A and C, $\times 12,000$; B, $\times 25,000$. D, electron microscopy of a thin section of cultured bovine corneal epithelial cells. $\times 17,500$.

Fig. 4. [³H]Thymidine autoradiography of corneal epithelial cells. Cells were plated and maintained as described in Ref. 12. The following additions were made: A, none (control); B, 10% calf serum; C, FGF at 100 ng/ml; D, EGF at 10 ng/ml. Arrows, labeled nuclei.

Fig. 5. Cross-section of control (A) and EGF-treated (B) cornea from bovine fetuses (crown-to-rump length, 15 cm) (3). The whole corneal epithelium exposed to EGF has this morphology. Cross-section of control (C) and FGF-treated (D) cornea from bovine fetuses (crown-to-rump length, 27 cm) (3). This represents an area which responded to FGF with an increased rate of proliferation. Other areas of the cornea stay as in C. $\times 100$.

Fig. 6. Effects of EGF and FGF on the growth of bovine corneal epithelial cells when maintained on different substrates. Corneal epithelial cells (2×10^4) were plated onto 6-cm tissue culture dishes which were either untreated (A to C) or coated with rabbit skin collagen (D to F), or which contained a Swiss 3T3 feeder layer (n/3, G to I). The cultures were maintained with 10% calf serum and Dulbecco modified Eagle's medium (A, D, and G) or with the addition of either EGF (100 ng/ml, B, E, and H) or FGF (100 ng/ml, F and I). The samples were added every other day, and the medium was renewed every 4 days. On Day 21 the dishes were fixed with 10% formalin and stained with rhodamine blue.

Fig. 7. Morphology of basal epithelial cells maintained on plastic, as opposed to those maintained on collagen-coated dishes. Corneal basal epithelial cells were maintained as described in Fig. 6 on collagen-coated dishes in the presence of Dulbecco modified Eagle's medium supplemented with 10% calf serum and EGF (25 ng/ml). The morphology of the cells is compared to that of the same cells maintained on plastic (A) under the same conditions. B represents the morphology of the culture after 2 days, while C, D, and E represent morphology after 7, 14, and 21 days, respectively. F represents a culture stained with rhodamine B. A to E are phase-contrast micrographs ($\times 100$). F is a bright-field photomicrograph ($\times 100$) of a 21-day-old culture fixed with 10% formalin and stained with rhodamine B. The keratinized layer stained intensely.

Fig. 8. Appearance of cloned adult bovine aortic endothelial cells maintained in tissue culture in the presence of FGF. A, monolayer of endothelial cells (13 passages, 65 generations) maintained in the presence of 10% calf serum and 100 ng of FGF per ml. The cells are polygonal, closely opposed, and have an indistinct border. Phase-contrast, $\times 150$. B, same monolayer as in A, but stained with silver nitrate to show the cell borders. $\times 300$. The cells showed the same organization as did preparations of endothelium stained *in situ*.

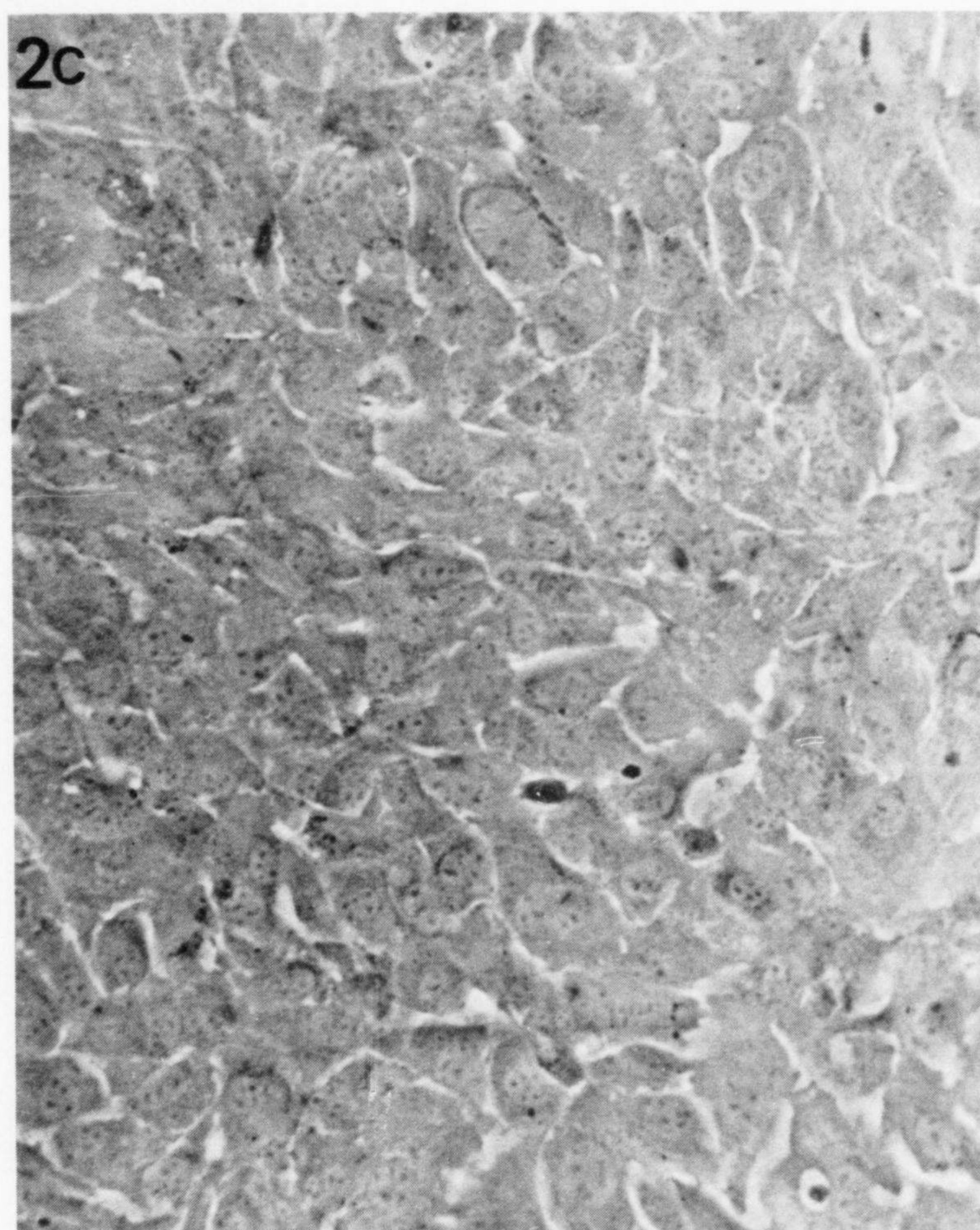
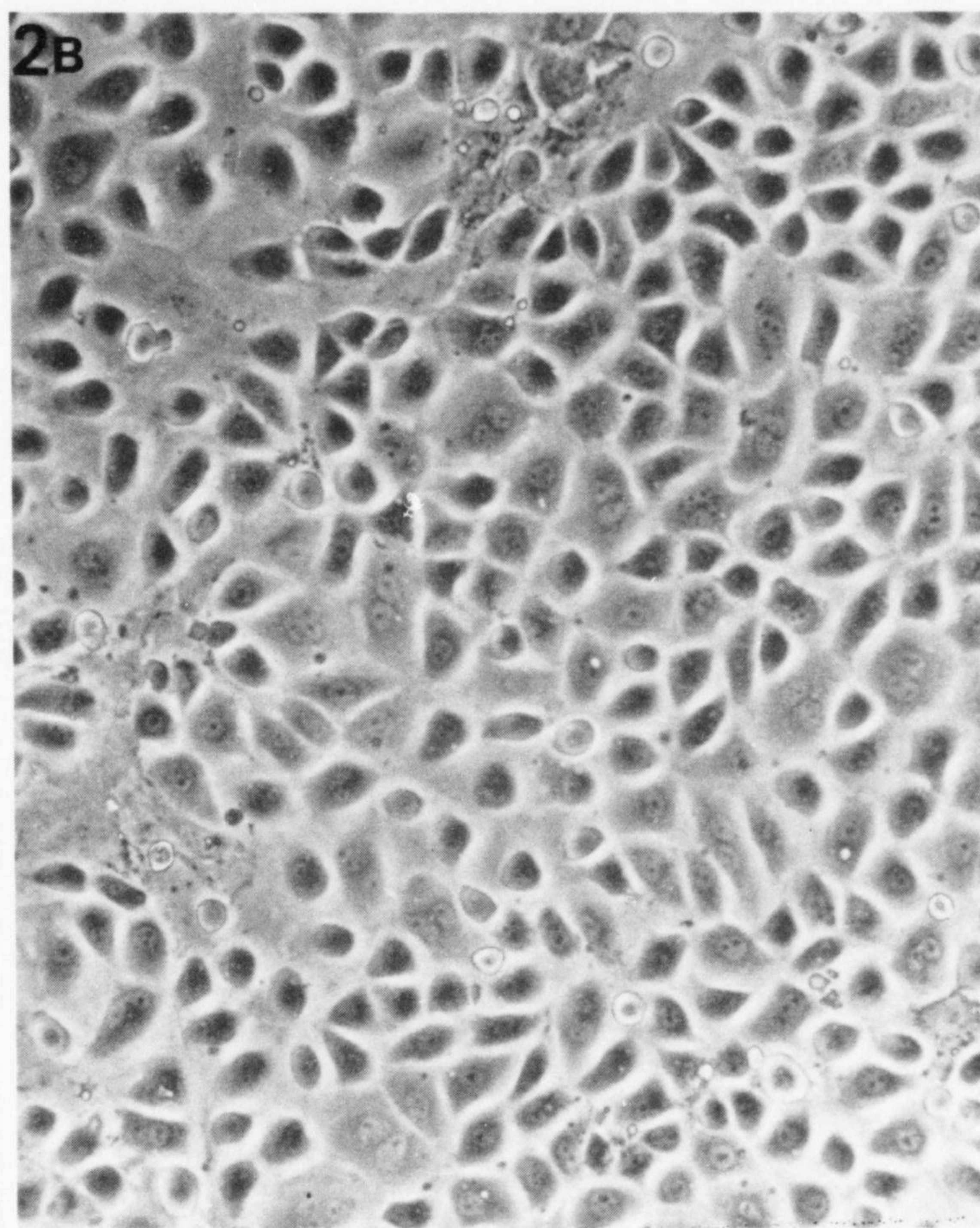
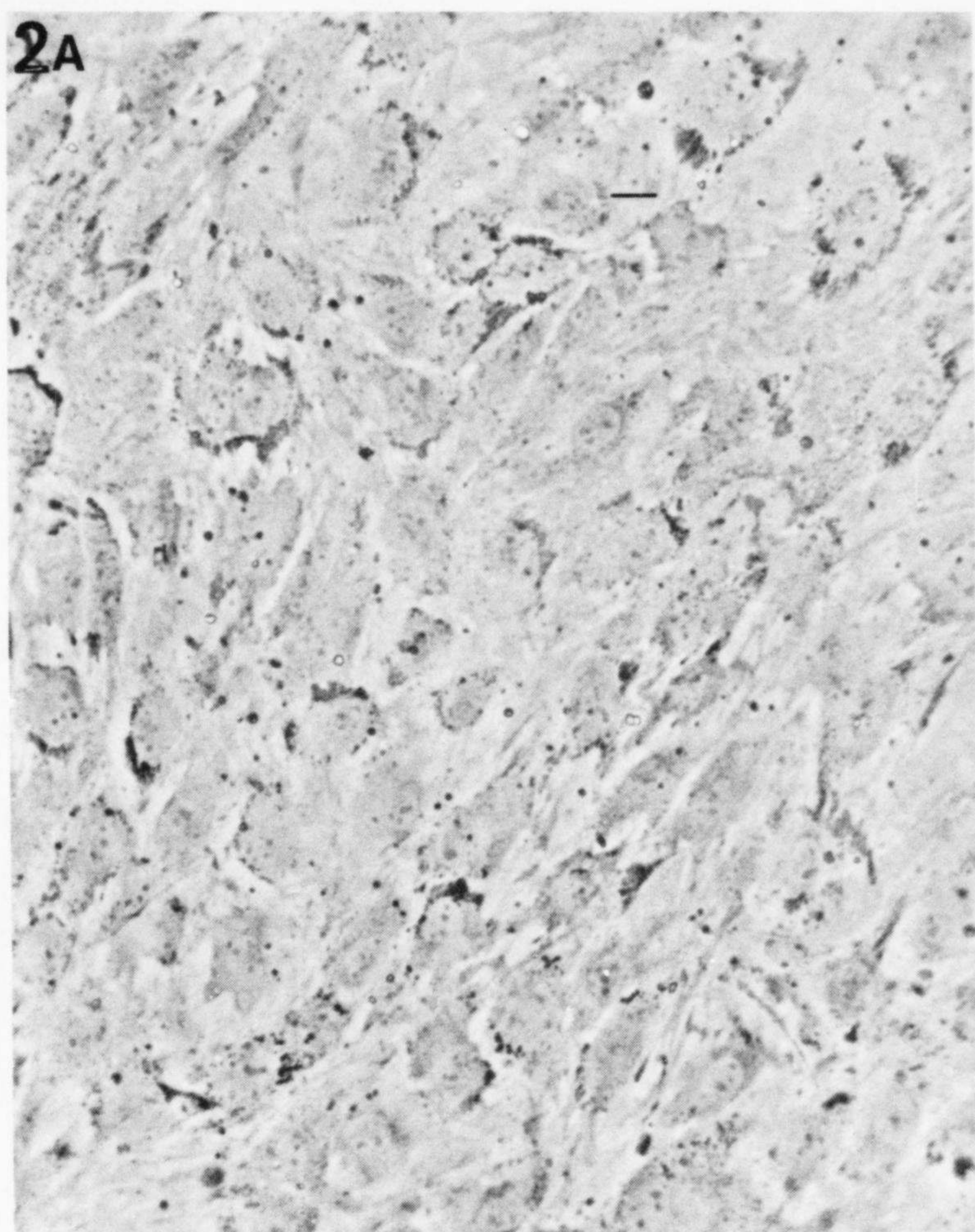
Fig. 9. Organization of corneal endothelial cells *in vivo* and *in vitro*. In A, segments of bovine cornea were removed from intact eyes, and the endothelial layer was stained with alizarin red. Inter-cellular boundary areas stain preferentially. In B, cultures of corneal endothelial cells were grown to confluence in medium containing 10% calf serum and 100 ng/ml of FGF and maintained for 14 days after reaching confluence. Inter-cellular boundary areas were stained *in vitro* with silver nitrate. The cells assume a polygonal shape both *in vitro* and *in vivo*.

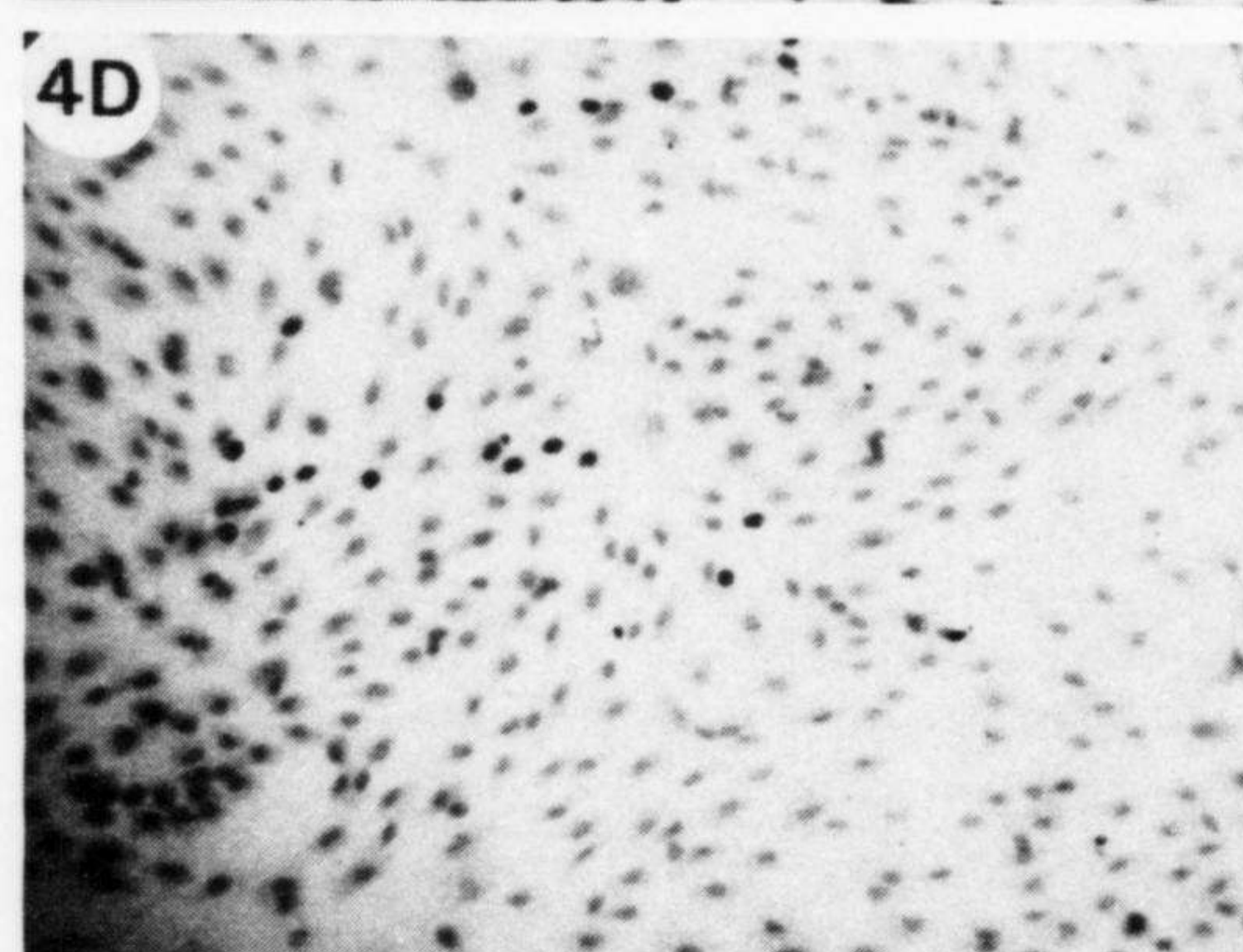
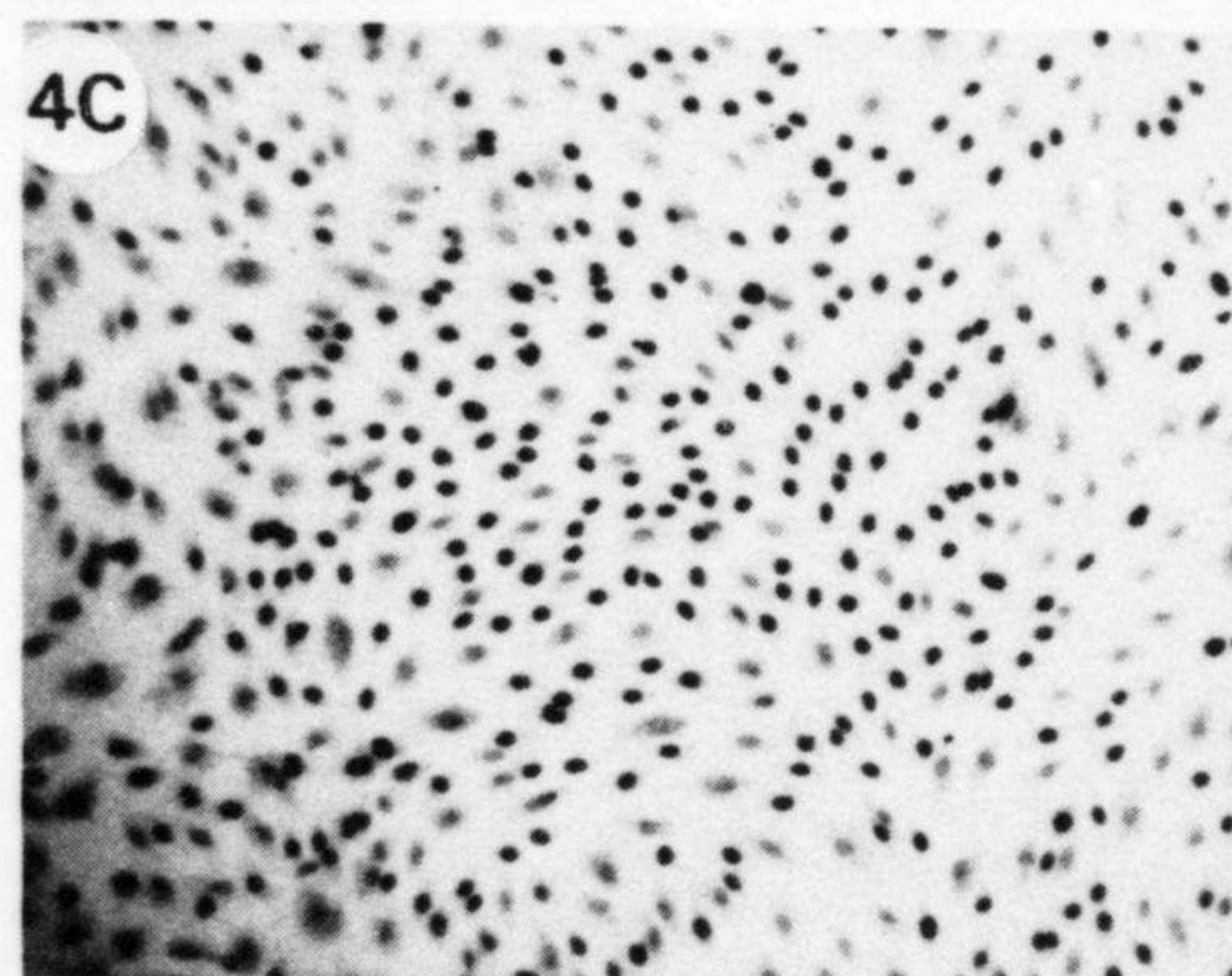
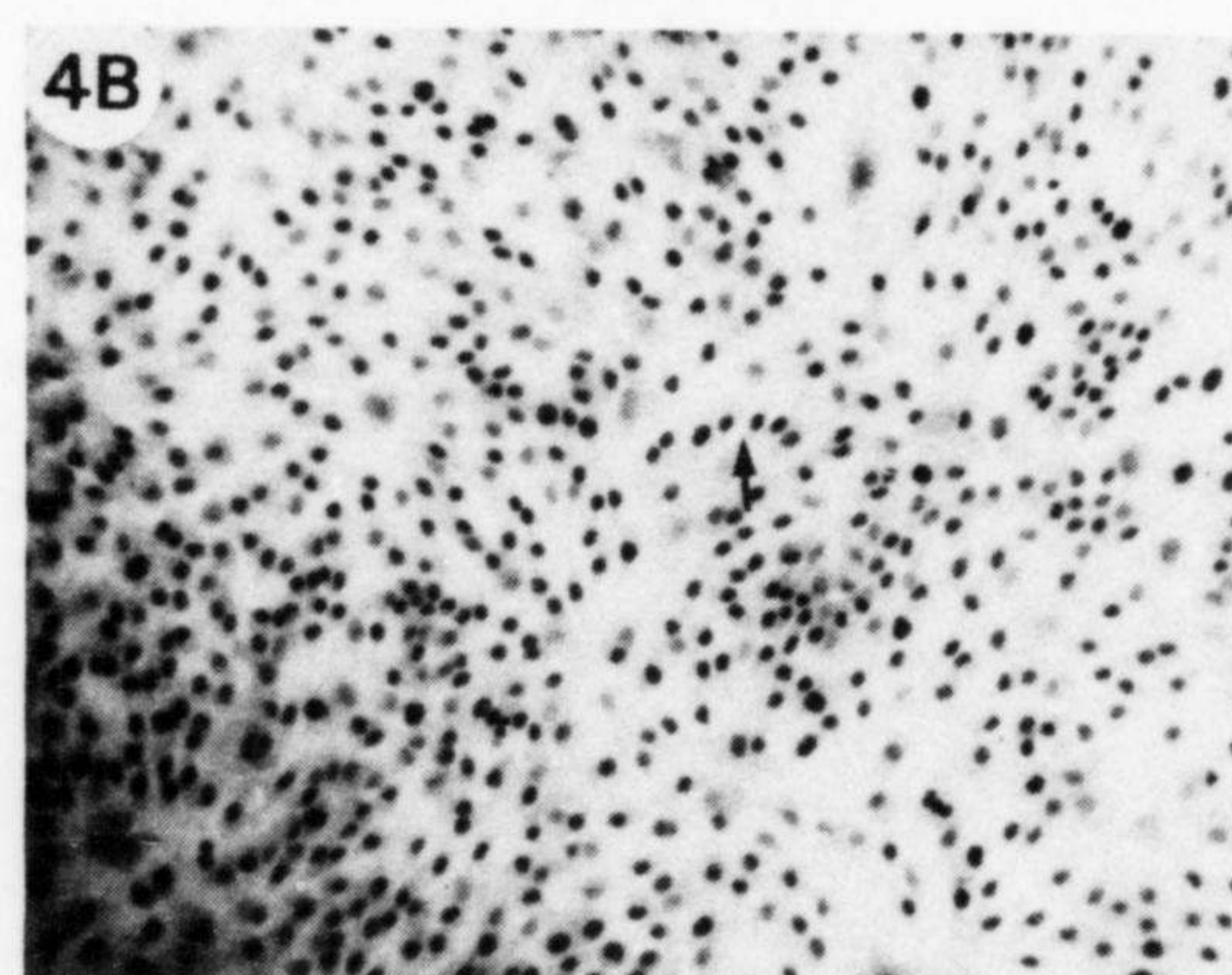
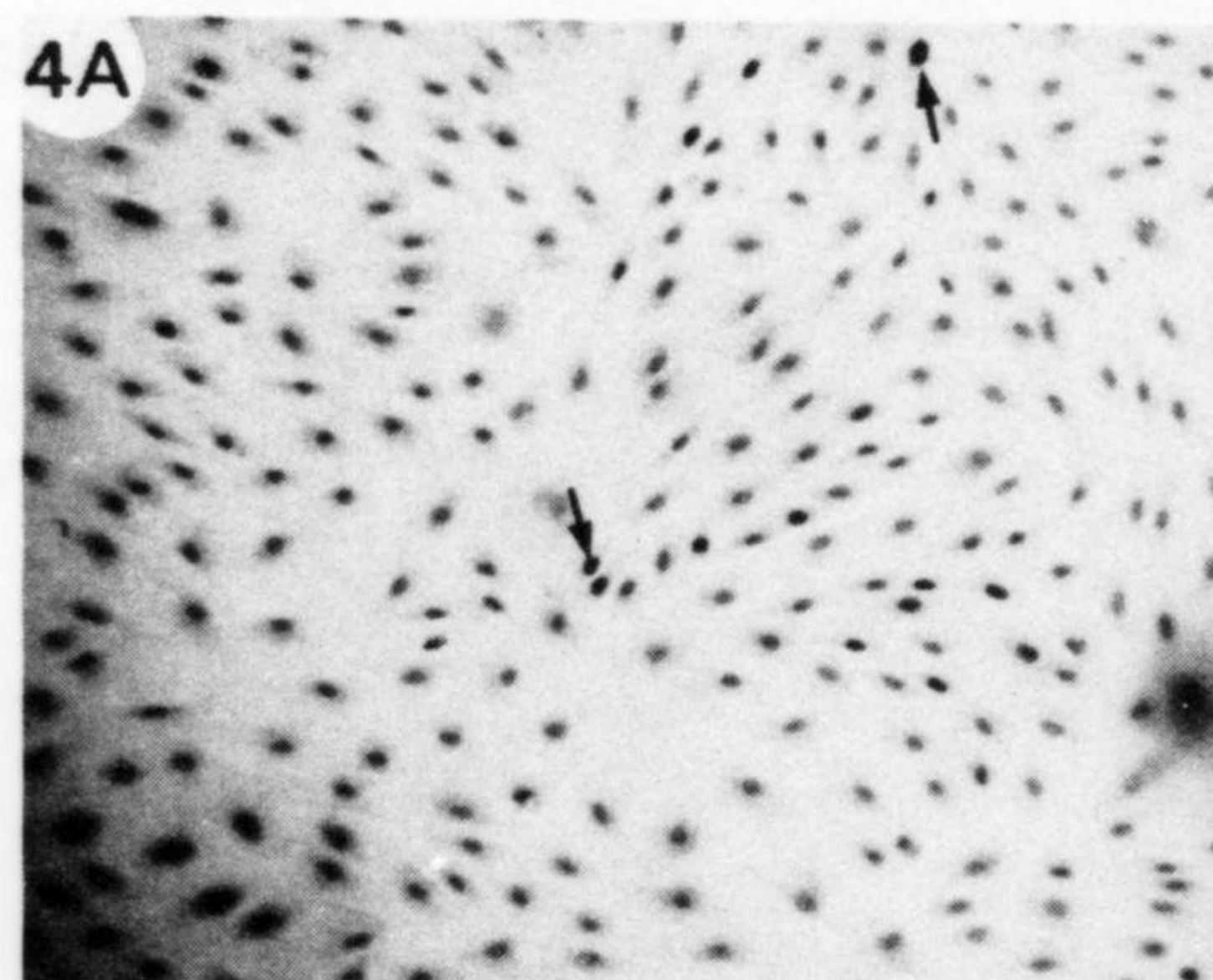
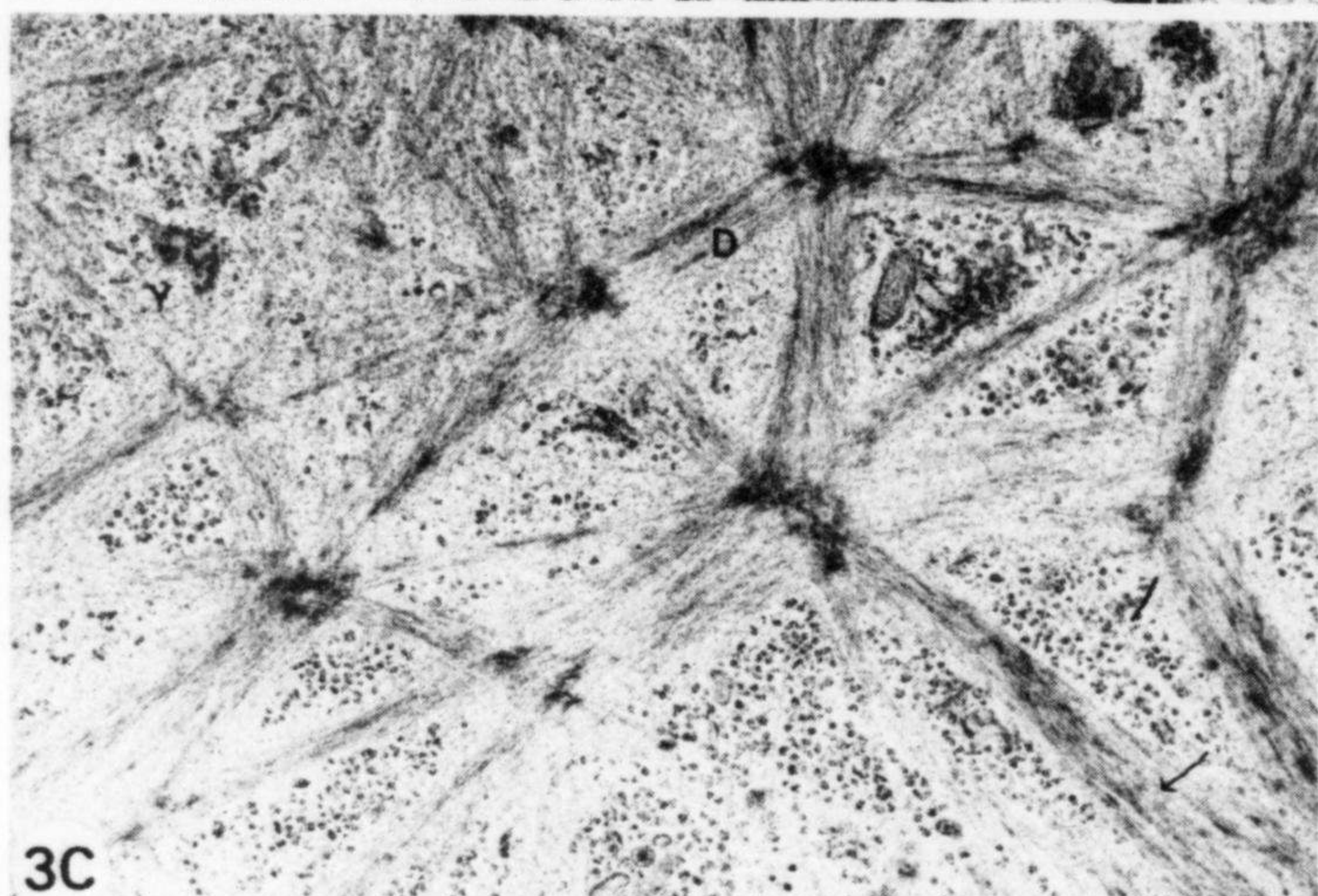
Fig. 10. A, confluent corneal endothelial cell cultures were treated with 1% Triton X-100 in phosphate-buffered saline to expose the basement membrane. The tissue cultures were then scratched with a needle to remove part of the basement membrane. B, zone of heavy deposit where the basement membrane is formed of multiple layers of collagenous material deposited on top of each other. C, indirect immunofluorescent localization of fibronectin associated in the basement membrane. Staining of the basement membrane revealed extensive labeling of the extracellular matrix.

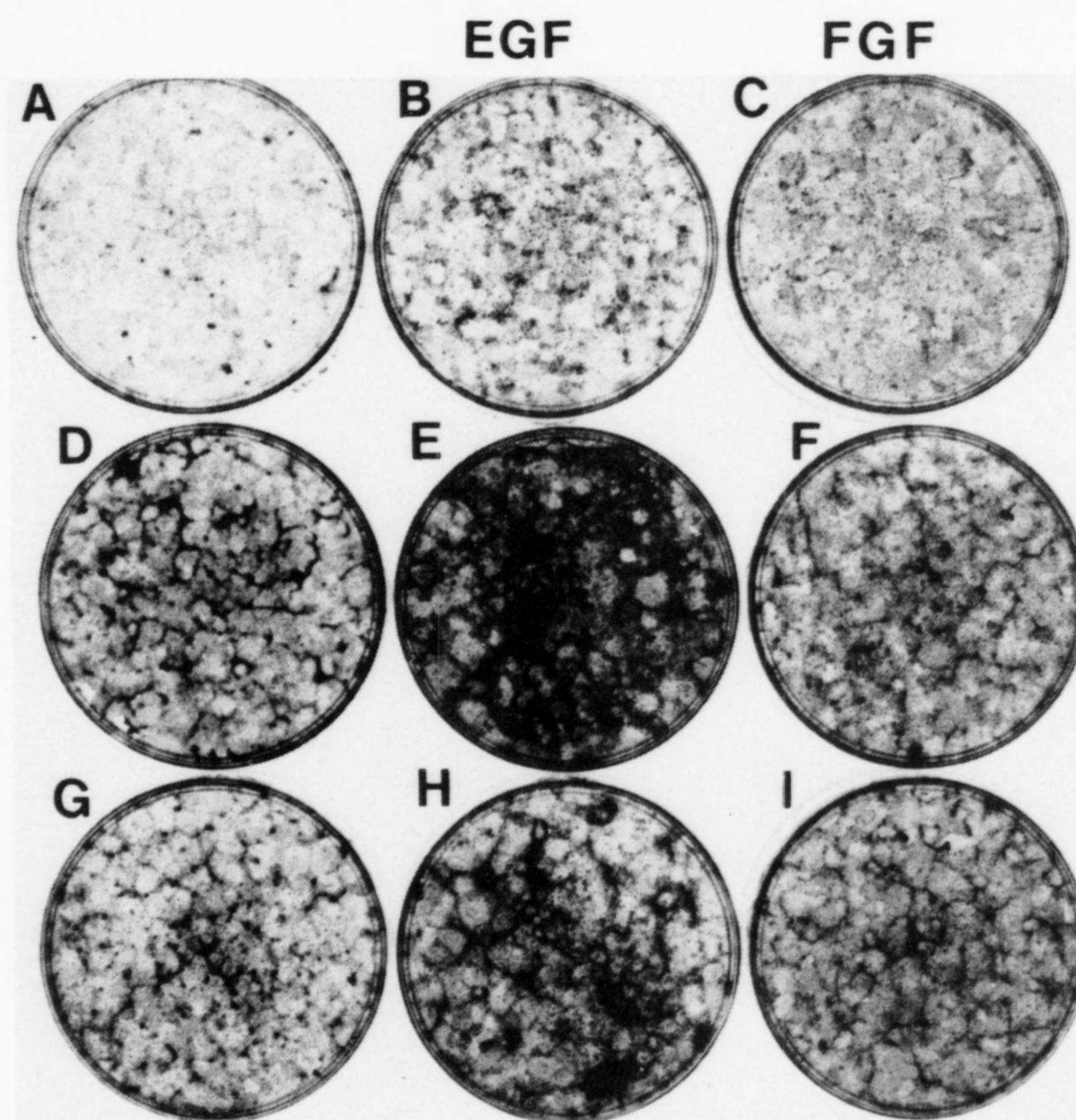
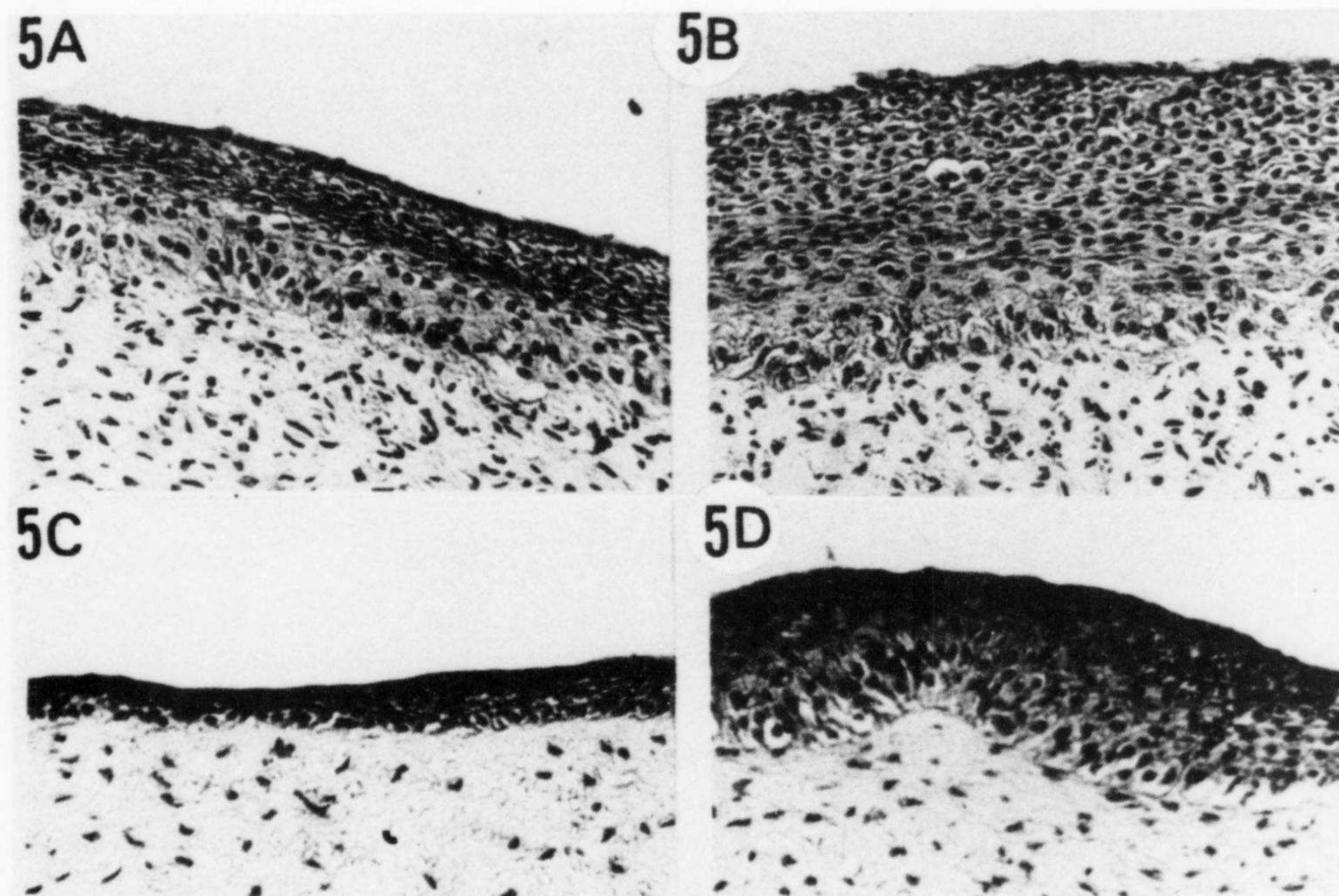
Fig. 11. Scanning electron micrograph of vascular endothelial cells and their basement membrane. A clone of adult bovine aortic endothelial (ABAE) cell cultures in their 20th passage (107 generations) was used. A, scanning electron micrograph of a 2-week-old ABAE culture treated briefly with 0.5% Triton X-100 in phosphate-buffered saline. This micrograph shows a single ABAE cytoskeleton and attached nucleus (n) stretched over the extracellular matrix (ecm). B, scanning electron micrograph of an ABAE monolayer treated very briefly with 0.5% Triton X-100. The extracellular matrix (ecm) is quite apparent, and many ABAE cytoskeletons are still attached to the substrate (3).

Fig. 12. Indirect immunofluorescence localization of fibronectin in vascular endothelial cells. A, phase-contrast micrograph of a 2-week-old culture after Triton X-100 treatment (3). B, treatment of the same field as in A with rabbit anti-fibronectin IgG, followed by staining with fluorescent goat anti-rabbit IgG, reveals intense labeling of extracellular matrix.

Fig. 13. Comparison of the ultrastructure of adult bovine aortic endothelial cells (6 passages, 42 generations) and vascular smooth muscle cells maintained in tissue culture. A, vascular endothelial cells. Numerous vesicles (v) are seen near the cell surface. Filaments (f, 60 to 70 Å and 100 Å) parallel to the plasma membrane are evident, as well as microtubules (m). n, nucleus. $\times 25,000$. B, vascular smooth muscle cells. The cytoplasm is filled with large bundles of myofilaments (mf), which contain fusiform dense bodies (arrows). n, nucleus. $\times 15,000$.







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